



The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MFM5220
Course title (English)	Algorithmic Investment
Course title (Chinese)	算法投资
Units	3
Description (English)	This course introduces mathematical models of portfolio theory; hands-on implementation of statistical test procedures for optimal portfolio construction and trading strategies.
Description (Chinese)	投资组合理论的数学模型；最优投资组合构建与交易策略的实践应用及统计检验方法。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

Stochastic processes, differential equations.



B. Co-requisites

C. Others

3. Learning outcomes

Students will be able to:

Understand the basics of financial products.

Understand the functioning of financial markets.

Understand the empirical financial data characteristics.

Understand major mathematical models for optimal execution.

Understand major mathematical models for market making.

Understand major mathematical models for pairs trading and statistical arbitrage.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

Introduction;

Preliminaries on financial products, markets, and trading;

Empirical and statistical evidences of financial data;

Stochastic control and optimal stopping;

Optimal execution;

Market making;

Pairs trading and statistical arbitrage.

5. Assessment scheme

Component/ method	% weight
Assignments	30
Midterm	20
Final project	50

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.



Type: **General**

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. **Feedback for evaluation**

Please specify forms of evaluation to generate feedback on classes from students.

Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

8. **Readings**

A.Required:



1. Algorithmic and high-frequency trading., 13: 9781107091146, Cartea, Álvaro, Sebastian Jaimungal, and José Penalva, Cambridge University Press, 2015

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Introduction
2	Preliminaries on financial products, markets, and trading
3	Empirical and statistical evidences of financial data
4	Stochastic control
5	Optimal stopping
6	Optimal execution (I)
7	Optimal execution (II)
8	Mid-Term Exam
9	Market making (I)
10	Market making (II)
11	Pairs trading and statistical arbitrage (I)
12	Pairs trading and statistical arbitrage (II)
13	Advanced topics on algorithmic trading (I)
14	Advanced topics on algorithmic trading (II)
15	Final project

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.



Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

15. Version date

Version number	1
As of (date)	2025/12/18